EXACT ENUMERATION / DATA / VERIFICATION

Rank-9 fusion rings

A verified dataset through multiplicity 2

Sébastien Palcoux

BIMSA

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Abstract

We document 2,225 verified based-isomorphism classes of fusion rings of rank 9 through multiplicity 2, including 2,083 at the maximum multiplicity. The producer reports a complete enumeration. The complete optimized source is supplied and hash-matched, but full enumeration has not been independently reproduced. The account separates the mathematical coverage argument, the completed search, and independent verification of the emitted tensors. All duality types are included, and no categorifiability filter is applied. Code, machine-readable counts, full multiplication tensors and provenance records accompany the text.

AI-generated research · independent verification pending. This manuscript, its code and data were developed with AI assistance. Independent human verification and Lean formalization are pending. Treat the results with caution: exact computational checks establish only their stated scope.

Scope. A census of based rings, not a classification of tensor categories. Only completed whole-rank results are released.

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1 Objects, equivalence and the counting convention

A *fusion ring* here is a free abelian group with a distinguished finite basis $B = \{b_0, b_1, \dots, b_{r-1}\}$, an associative multiplication with unit b_0, and an involution $i \mapsto i^*$ fixing 0. Its structure constants satisfy

$$b_i b_j = \sum_k N_{ij}^k b_k, \qquad N_{ij}^k \in \mathbb{Z}_{\geq 0},$$

$$N_{0j}^k = N_{j0}^k = \delta_{jk}, \qquad N_{ij}^0 = \delta_{i^*,j}, \tag{1}$$

$$N_{ij}^k = N_{i^*k}^j = N_{kj^*}^i. \tag{2}$$

These conventions agree with the usual based-ring definition; see [1, 2]. No assumption of categorifiability, commutativity or integrality of Frobenius–Perron dimensions is imposed unless proved in the relevant rank.

A based isomorphism is a bijection of the distinguished bases preserving the unit and multiplication. In tensor notation it is a permutation q fixing 0 such that

$$N_{ij}^k = N'^{q(k)}_{q(i)q(j)} \quad \text{for every } i, j, k.$$

The coefficient of the unit determines the involution, so such an isomorphism automatically intertwines duality. The *multiplicity* is the maximum entry of the *full* ordered multiplication tensor. It is at least one, including for group rings. Put

$$c_r(m) = \#\{\text{rank-}r \text{ fusion rings of multiplicity exactly } m\}/\cong, \qquad C_r(M) = \sum_{m=1}^{M} c_r(m).$$

A list through M contains all rings of multiplicity at most M, not only those of multiplicity equal to M. This distinction is essential when exporting OEIS b-files.

1.1 Why duality types can be searched separately

An involution on the $r-1$ nonunit elements consists of p disjoint transpositions and $r-1-2p$ fixed points, with $0 \leq p \leq \lfloor (r-1)/2 \rfloor$. Fix one representative d_p of each type. An isomorphism between two tensors with involution d_p commutes with d_p. Hence it is sufficient to quotient that stratum by the centralizer

$$Z_{S_{r-1}}(d_p), \qquad |Z_{S_{r-1}}(d_p)| = 2^p p!(r-1-2p)!.$$

Different p cannot be isomorphic. Therefore a whole-rank census is complete exactly when every required stratum has been searched exhaustively. Finishing only the non-self-dual strata does not certify a whole-rank counting term.

2 The certified release at rank 9

The released bound is $M = 2$ and the cumulative number of classes is $C_9(2) = 2,225$. Every duality type is included. The accompanying full tensor file has one record per based-isomorphism class.

Exact multiplicity m	$c_9(m)$
1	142
2	2,083

The data and their checksums are recorded in `results/census.json`. The corresponding completed-run evidence and independent audit are under `verification/rank9/`.

3 Supplied high-rank source and independent checks

The complete optimized enumeration source and completed producer-run records are retained under `verification/engine_20261023/` and the rank-specific verification directory. The run completed every duality stratum without a clock limit. The supplied tensors preserve their original physical line order; a categorical label never changes that input identifier.

Fresh independent integration checks establish all tensor axioms, exact based-isomorphism uniqueness, the counts and both compressed and uncompressed checksums. The optimized uniqueness checker uses exact individualization and refinement and checks every returned isomorphism witness against the complete tensor. Exhaustive permutation mode remains available. Parameter re-export is byte-identical. These checks establish soundness and uniqueness of the supplied output; they do not replace independent repetition of the entire search.

Categorification in the companion project. The separate rank-nine multiplicity-two manuscript records 2,043 general complex exclusions, 36 unitary models and 4 unresolved rings among the 2,083 exact-multiplicity-two inputs. The 142 multiplicity-one rings retain 102 exclusions and 40 models. Thus the cumulative rank-nine classification through multiplicity two has 2,145 exclusions, 76 models and 4 unresolved cases. These mathematical decisions are separate from the census-completeness qualification. The current subring criterion requires an explicit negative proof; omission from a positive model catalog is never an exclusion. The accepted ledger, exact proofs and replay archive bind all model permutations and exclusion dependencies to the source tensors. The multiplicity-one source may use different bases: all earlier IDs are preserved through explicit based transports, with fourteen nonidentity maps.

4 An exhaustive finite-domain model

4.1 Coordinates directly from reciprocity

On nonunit triples apply the two transformations

$$(i,j,k) \longmapsto (i^*,k,j), \qquad (i,j,k) \longmapsto (k,j^*,i).$$

Assign one variable $x_O \in \{0,\dots,M\}$ to each resulting orbit O. The unit entries and N_{ij}^0 are already fixed. Thus every tensor satisfying unit, duality and reciprocity occurs exactly once as an assignment of orbit coordinates. Three successive reciprocity transformations give $N_{ij}^k = N_{j^*i^*}^{k^*}$, so duality is an anti-involution. In particular, the self-dual stratum is commutative, at every rank.

For every nonunit i,j,k,ℓ impose the integer polynomial

$$\sum_s N_{ij}^s N_{sk}^\ell - \sum_s N_{jk}^s N_{is}^\ell = 0. \tag{3}$$

Associativity involving a unit is automatic. When $\ell = 0$, the two sides reduce to $N_{ij}^{k^*}$ and $N_{jk}^{i^*}$, which agree by reciprocity and the anti-involution. Therefore these generated polynomials are necessary and sufficient for associativity. Removing zero polynomials and normalizing signs of repeated equations changes no solution.

4.2 Safe pruning, including singular linear systems

The baseline solver starts with a finite domain for every variable. It evaluates an enclosing interval for each polynomial and rejects a branch only if that interval excludes zero. If an equation contains at most two unfixed variables, it tests every pair of values in their current domains and removes unsupported values. These operations preserve every solution.

Equations that have become linear are row-reduced over $\mathbb{F}_{257}$. An inconsistent modular system cannot have an integer solution. Consequences involving one or two variables restrict their finite domains. Rank deficiency by itself never rejects a branch: free variables remain in the exhaustive search. Since $M \le 15 < 257$, a residue determines at most one permitted integer. The final leaf is always checked against every original integer polynomial.

Lexicographic comparison with the duality centralizer retains the least coordinate vector in each orbit. A comparison stops at its first undecided coordinate; one cannot infer a later lexicographic inequality past an earlier unknown entry. Any extra prefix inequalities are necessary conditions for the same least representative.

Theorem 4.1 (Conditional completeness of a terminated search)**.** *Fix r, M and a duality type. If the finite-domain solver terminates normally and processes every branch of its search tree, it emits exactly one tensor from each based-isomorphism class satisfying that type and bound. If every duality type terminates normally, their union is the complete whole-rank census.*

Proof. Every admissible tensor has an orbit-coordinate vector in the initial box. Each pruning rule is a necessary condition, so the least representative of that tensor is never deleted. At each unpruned node the first unfixed variable is assigned every remaining value. Induction on the number of unfixed variables therefore visits each feasible least vector. Leaf tests prove soundness, and lexicographic minimality proves uniqueness. Finally, duality types are disjoint and exhaust all possibilities. □

A parallel run adaptively splits prefix subtrees into every bounded value of the next unfixed coordinate. The resulting frontier subtrees are disjoint and their union is the original tree. A run is marked complete only after every frontier subtree finishes. A time limit is not a mathematical rejection rule: a timeout leaves the larger census uncertified.

5 Self-dual reconstruction from one multiplication matrix

Put $n = r - 2$. Choosing a nonunit element gives a symmetric matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & a & u^{\mathsf{T}} \\ 0 & u & Q \end{pmatrix}, \qquad u \in \mathbb{Z}^n_{\ge 0}, \quad Q = Q^{\mathsf{T}} \in M_n(\mathbb{Z}_{\ge 0}).$$

The remaining parameters are a symmetric cubic tensor on the n tail indices. Relabelling allows u to be nondecreasing. If $b = u_1$ is the least mixed repeated-index coefficient, then every such coefficient is at least b; in particular $Q_{ii} \ge b$. Ties in u allow the corresponding diagonal entries of Q to be sorted. These are necessary conditions on an orbit minimum, not extra axioms imposed on the ring.

5.1 The two contraction identities

Let R_i be the tail block of multiplication by the ith tail basis element. Comparing the lower blocks of its commutator with A gives

$$R_i u = (Q^2 + uu^{\mathsf{T}} - I - aQ)e_i, \tag{4}$$

$$R_i Q = QR_i - (Qe_i)u^{\mathsf{T}} + u(Qe_i)^{\mathsf{T}}. \tag{5}$$

Writing $s = \sum_j u_j$, each contraction lies in $[0, Ms]$ and is divisible by $\gcd(u_1, \dots, u_n)$, with zero required if $u = 0$. Mixed-coefficient bounds strengthen the diagonal lower bound to $b(s - u_i)$ and the off-diagonal lower bound to $b(u_i + u_j)$. Partial sums of squares in Q^2 are enclosed by letting each missing entry range over $[0, M]$. These necessary interval and divisibility conditions eliminate many first matrices before cubic completion.

5.2 A finite-field reconstruction with an exhaustive fallback

Define

$$V_k = Q^k u, \quad K = [V_0, \dots, V_{n-1}], \quad H_0 = Q^2 + uu^{\mathsf{T}} - I - aQ,$$
$$H_{k+1} = QH_k - (u^{\mathsf{T}} V_k)Q + uV_{k+1}^{\mathsf{T}}.$$

Induction using (5) proves $R_i V_k = H_k e_i$. Each H_k is affine in a. If K is invertible modulo a prime $p > M$, then

$$R_i = [H_0 e_i, \dots, H_{n-1} e_i] K^{-1} \pmod{p}.$$

Each reconstructed residue has at most one representative in $[0, M]$. Retaining only those representatives and then checking all original integer equations gives an exact reconstruction, not an approximate numerical method.

The implementation tries $p = 257$ and $p = 17$. A preliminary row system can reject values of a without reconstructing all tail entries. If either system has free coordinates, its compatibility equations are retained. With one or two free coordinates, all their permitted values are tested. With more free coordinates the general exhaustive completion is used. If neither prime gives an invertible K, the solver searches every surviving integer value of a and every compatible remaining coordinate. In particular, an integer matrix that happens to be singular modulo a chosen prime is never lost.

5.3 Folding a fixed invertible element

When $u = 0$, $a = 0$ and $Q = I$, the chosen element g satisfies $g^2 = 1$ and $gX_i = X_i$ for all tail elements. The tail product then has the form

$$X_i X_j = \delta_{ij}(1 + g) + \sum_k t_{ijk} X_k.$$

Its associativity equations are the rank-$(r-1)$ symmetric equations with the unit contribution changed from one to two. This is an auxiliary weighted system, not a new fusion ring. Replace I by $2I$ in the contraction formulae, enumerate that smaller system, and unfold. Each unfolded tensor is minimized under the original relabellings and checked against the original equations.

The distinguished pair $\{1, g\}$ is intrinsic: another invertible element h fixed by g would yield $g = 1$ after multiplying $gh = h$ by h^{-1}. Thus folding introduces no hidden identifications. It is used only in this proved case; weighted systems are not recursively folded under an unproved extension of the argument.

6 What the independent verification establishes

The enumerator and verifier have separate mathematical tasks. Exhaustiveness uses the complete coordinate parameterization, the safe-pruning proof and the normal completion of every required branch. The verifier reads only the emitted *full tensors*; it neither reconstructs the parameterization nor trusts the generator's polynomial list.

For every tensor it checks nonnegative integral coefficients, both unit laws, uniqueness and involutivity of the dual, both reciprocity identities, and every integer associativity identity for all four indices. It then canonicalizes under *every* unit-fixing basis permutation. A matching hash is only a lookup aid: actual canonical tensors are compared. This detects both literal repetitions and different labellings of an isomorphic based ring.

Passing these checks proves soundness and uniqueness of the emitted data; by itself it would not prove that no ring was omitted. That is why normal completion logs and the mathematical

coverage argument are retained separately. Comparison with the original ancillary file is an additional containment/regression test, not an input to enumeration and not a replacement for an exhaustiveness proof.

6.1 Provenance and reproducibility

The retained Vercleyen–Slingerland v4 ancillary file has SHA-256

`b587c3f683157369da7cc090f762bbff0e18dcf11acc3f5031d6bc0c643843f8`

Its 28,451 records contain 25,138 distinct classes. The 3,313 redundant records occur at rank five and exact multiplicities 9–12. Their original and corrected counts are respectively

$$(1463, 1794, 2283, 3049) \quad \longrightarrow \quad (863, 1082, 1383, 1948).$$

Every deletion has an explicit basis-permutation witness in the audit folder. This audit does not establish completeness of the source's partial-search cells.

The primary manifest `results/census.json` records every released count, bound, filename and checksum. Compressed tensors use the original line-per-ring, nested-curly-brace format. An entry is $N[i][j][k] = N_{ij}^k$: the output index has already been dualized where the internal parameter convention requires it. The raw tensor needs no additional index conversion.

6.2 Certified datasets and reproducibility

The primary manifest gives the complete bound, exact counts and data checksums for every rank. Each `verification/rankR/` directory records the independent audit and source identity for that dataset. For the laptop censuses it also retains the completed enumeration, commands, environment, phase timings and nonempty subprocess logs. Every required duality stratum completed normally; independent canonicalization found zero duplicate classes.

Independent checks use the supplied tensors; they do not rerun enumeration. The per-rank reports identify the verification environment and source revision, and certify agreement with the established exact-multiplicity count prefix. Hosted audits and local checks are identified separately in `verification/README.md`; see `docs/PROVENANCE.md` for attribution.

The local launcher has no clock limit, while the manual hosted workflow uses a finite shared deadline covering every computational phase. A timeout never certifies the unfinished bound. Only a complete, independently verified whole-rank candidate can enter the release; a later failed attempt cannot replace it. Reproduction commands and supported limits are in `docs/REPRODUCIBILITY.md`.

7 Reading and reusing the data

The result files count based rings, not monoidal categories or realizations of a ring. One tensor can admit several inequivalent categorifications, or none. No number here is a classification of fusion categories. The rank-specific count files use exact multiplicity; cumulative totals belong only in the summary column. Proposed OEIS additions must be regenerated from a complete manifest, not copied from an interrupted log.

The retained code was generated with AI assistance and is accompanied by explicit mathematical coverage arguments and independent integer checks. These checks are not a formal proof in a proof assistant. The present text is an explanatory computational companion, not a claim of journal acceptance or external peer review.

The completed-search and source qualification in Section 2 applies throughout: the complete optimized source is supplied and hash-matched, but independent output checks do not replace a fresh exhaustive enumeration.

8 Reproduction commands

From the repository root on a local computer, request a complete run with no clock limit. An interruption does not certify a smaller census.

```
python3 scripts/run_laptop.py --rank 9 --bound 2
```

To verify the supplied release without recomputing it:

```
python3 scripts/check_release.py --full
```

To verify only this released rank, use `python3 scripts/check_release.py -ranks 9 -full`. Reproduction and tensor verification are distinct operations.

References

[1] P. Etingof, S. Gelaki, D. Nikshych and V. Ostrik, *Tensor Categories*, Mathematical Surveys and Monographs 205, American Mathematical Society, 2015.

[2] G. Vercleyen and J. Slingerland, *On Low Rank Fusion Rings*, Journal of Mathematical Physics 64 (2023), 091703; arXiv:2205.15637v4.

[3] V. Ostrik, *On formal codegrees of fusion categories*, Mathematical Research Letters 16 (2009), 895–901; arXiv:0810.3242.

[4] J. Dong and S. Palcoux, rank-four and rank-five census derivations, project manuscripts and computational supplements, September 2026. The relevant derivations are retained in the repository's `methods/` directory.

[5] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entries A348305, A354471, A354472, A354473, A354475, A354476 and A354477. Proposed updates in this repository are not submitted edits.